Unit 15: Related Rates — Full Solutions

Warm-up

1. Circle, $A = \pi r^2$, $\dfrac{dr}{dt} = 3$, find $\dfrac{dA}{dt}$ at $r = 5$.

Differentiate with respect to t:

$$\frac{dA}{dt} = 2\pi r \cdot \frac{dr}{dt}$$

Plug in $r = 5$, $\dfrac{dr}{dt} = 3$:

$$\frac{dA}{dt} = 2\pi(5)(3) = 30\pi \text{ cm}^2/\text{s}$$

2. Square, $A = s^2$, $\dfrac{ds}{dt} = 2$, find $\dfrac{dA}{dt}$ at $s = 4$.

$$\frac{dA}{dt} = 2s \cdot \frac{ds}{dt} = 2(4)(2) = 16 \text{ cm}^2/\text{s}$$

3. Sphere, $V = \dfrac{4}{3}\pi r^3$, $\dfrac{dr}{dt} = 4$, find $\dfrac{dV}{dt}$ at $r = 3$.

$$\frac{dV}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$$

$$\frac{dV}{dt} = 4\pi(3)^2(4) = 4\pi(9)(4) = 144\pi \text{ cm}^3/\text{s}$$

4. Cube, $V = s^3$, $\dfrac{ds}{dt} = 1$, find $\dfrac{dV}{dt}$ at $s = 5$.

$$\frac{dV}{dt} = 3s^2 \cdot \frac{ds}{dt} = 3(25)(1) = 75 \text{ cm}^3/\text{s}$$

5. Circle, $A = \pi r^2$, $\dfrac{dr}{dt} = -2$ (shrinking), find $\dfrac{dA}{dt}$ at $r = 6$.

$$\frac{dA}{dt} = 2\pi r \cdot \frac{dr}{dt} = 2\pi(6)(-2) = -24\pi \text{ cm}^2/\text{s}$$

The negative sign confirms the area is shrinking, at a rate of 24π cm²/s.

6. Equilateral triangle, $A = \dfrac{\sqrt{3}}{4}s^2$, $\dfrac{ds}{dt} = 3$, find $\dfrac{dA}{dt}$ at $s = 6$.

$$\frac{dA}{dt} = \frac{\sqrt{3}}{4} \cdot 2s \cdot \frac{ds}{dt} = \frac{\sqrt{3}}{2}s \cdot \frac{ds}{dt}$$

$$\frac{dA}{dt} = \frac{\sqrt{3}}{2}(6)(3) = 9\sqrt{3}\ \text{cm}^2/\text{s}$$

Standard

7. Ladder: let x = distance of bottom from wall, y = height of top on wall. $x^2 + y^2 = 10^2 = 100$.

Given $\frac{dx}{dt} = 2$ ft/s. Find $\frac{dy}{dt}$ when $x = 6$.

First find y when $x = 6$: $y^2 = 100 - 36 = 64 \Rightarrow y = 8$.

Differentiate the connecting equation with respect to t:

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

Solve for $\frac{dy}{dt}$:

$$\frac{dy}{dt} = -\frac{x}{y} \cdot \frac{dx}{dt} = -\frac{6}{8}(2) = -\frac{3}{2}\ \text{ft/s}$$

Answer: the top is sliding down at $\frac{3}{2}$ ft/s (the negative sign means y is decreasing).

8. Balloon: let h = height, z = distance from observer. Fixed horizontal distance 100 ft, so $z^2 = 100^2 + h^2 = 10000 + h^2$.

Given $\frac{dh}{dt} = 5$ ft/s. Find $\frac{dz}{dt}$ when $h = 75$.

First find z: $z^2 = 10000 + 75^2 = 10000 + 5625 = 15625 \Rightarrow z = 125$ (since $125^2 = 15625$).

Differentiate:

$$2z\frac{dz}{dt} = 2h\frac{dh}{dt}$$

$$\frac{dz}{dt} = \frac{h}{z} \cdot \frac{dh}{dt} = \frac{75}{125}(5) = \frac{375}{125} = 3\ \text{ft/s}$$

9. Cylinder, fixed radius $r = 3$ ft. $V = \pi r^2 h = 9\pi h$ (since r never changes).

Given $\frac{dV}{dt} = 2$ ft³/min. Differentiate:

$$\frac{dV}{dt} = 9\pi\frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{2}{9\pi}\ \text{ft/min}$$

10. Two cars: let x = eastbound car's distance, y = northbound car's distance, z = distance between them. $z^2 = x^2 + y^2$.

Given $\frac{dx}{dt} = 40$ mph, $\frac{dy}{dt} = 30$ mph. Find $\frac{dz}{dt}$ when $x = 3$, $y = 4$.

First find z: $z^2 = 9 + 16 = 25 \Rightarrow z = 5$.

Differentiate:

$$2z\frac{dz}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}$$

$$\frac{dz}{dt} = \frac{x\frac{dx}{dt} + y\frac{dy}{dt}}{z} = \frac{(3)(40) + (4)(30)}{5} = \frac{120 + 120}{5} = \frac{240}{5} = 48 \text{ mph}$$

11. Rectangle, $A = LW$. Given $\frac{dL}{dt} = 3$ cm/s, $\frac{dW}{dt} = -2$ cm/s. Find $\frac{dA}{dt}$ at $L = 10$, $W = 6$.

Differentiate using the product rule:

$$\frac{dA}{dt} = \frac{dL}{dt} \cdot W + L \cdot \frac{dW}{dt}$$

$$\frac{dA}{dt} = (3)(6) + (10)(-2) = 18 - 20 = -2 \text{ cm}^2/\text{s}$$

The area is currently decreasing, at a rate of 2 cm^2/s.

12. Shadow: let x = person's distance from the lamp post, s = shadow length. By similar triangles (comparing the lamp post's height/distance triangle to the person's height/shadow triangle):

$$\frac{15}{x+s} = \frac{6}{s}$$

Cross-multiply: $15s = 6(x+s) = 6x + 6s$, so $9s = 6x$, giving $s = \frac{2}{3}x$.

Differentiate this simplified relationship directly:

$$\frac{ds}{dt} = \frac{2}{3} \cdot \frac{dx}{dt} = \frac{2}{3}(5) = \frac{10}{3} \text{ ft/s}$$

The shadow length is growing at $\frac{10}{3}$ ft/s — notice this rate stays constant no matter how far the person has already walked.

13. Rectangle, fixed width $W = 4$, length L growing at $\frac{dL}{dt} = 2$ cm/s. Diagonal: $d^2 = L^2 + W^2 = L^2 + 16$.

Find $\frac{dd}{dt}$ when $L = 3$: first find d: $d^2 = 9 + 16 = 25 \Rightarrow d = 5$.

Differentiate:

$$2d\frac{dd}{dt} = 2L\frac{dL}{dt}$$

$$\frac{dd}{dt} = \frac{L}{d} \cdot \frac{dL}{dt} = \frac{3}{5}(2) = \frac{6}{5} = 1.2 \text{ cm/s}$$

Challenge

14. Conical tank: top radius $R = 4$ ft, height $H = 10$ ft (fixed tank dimensions). Let r and h be the radius and height of the water's surface at any moment.

By similar triangles, $\frac{r}{h} = \frac{R}{H} = \frac{4}{10} = \frac{2}{5}$, so $r = \frac{2}{5}h$.

Substitute into the cone volume formula to eliminate r:

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\left(\frac{2}{5}h\right)^2 h = \frac{1}{3}\pi \cdot \frac{4}{25}h^2 \cdot h = \frac{4\pi}{75}h^3$$

Differentiate with respect to t:

$$\frac{dV}{dt} = \frac{4\pi}{75} \cdot 3h^2 \cdot \frac{dh}{dt} = \frac{4\pi}{25}h^2 \cdot \frac{dh}{dt}$$

Given $\frac{dV}{dt} = -3$ ft^3/min (draining, so negative) and $h = 5$:

$$-3 = \frac{4\pi}{25}(25) \cdot \frac{dh}{dt} = 4\pi \cdot \frac{dh}{dt}$$

$$\frac{dh}{dt} = -\frac{3}{4\pi} \text{ ft/min}$$

The water level is dropping at a rate of $\frac{3}{4\pi}$ ft/min.

15. Ships: let x = remaining east-west gap between the ships, y = Ship B's northward distance, z = distance between the ships.

Ship A starts 100 km west of Ship B and sails east at 15 km/h, closing the east-west gap, so $x = 100 - 15t$. Ship B sails north at 20 km/h, so $y = 20t$.

At $t = 4$ (4:00 PM): $x = 100 - 15(4) = 100 - 60 = 40$. $y = 20(4) = 80$.

$$z^2 = x^2 + y^2 = 40^2 + 80^2 = 1600 + 6400 = 8000$$

$$z = \sqrt{8000} = 40\sqrt{5}$$

Rates: $\dfrac{dx}{dt} = -15$ (the east-west gap is shrinking), $\dfrac{dy}{dt} = 20$.

Differentiate $z^2 = x^2 + y^2$:

$$2z\frac{dz}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}$$

$$\frac{dz}{dt} = \frac{x\frac{dx}{dt} + y\frac{dy}{dt}}{z} = \frac{(40)(-15)+(80)(20)}{40\sqrt{5}} = \frac{-600+1600}{40\sqrt{5}} = \frac{1000}{40\sqrt{5}} = \frac{25}{\sqrt{5}}$$

Rationalize: $\dfrac{25}{\sqrt{5}} = \dfrac{25\sqrt{5}}{5} = 5\sqrt{5}$.

Answer: the distance between the ships is increasing at a rate of $5\sqrt{5} \approx 11.18$ km/h.

16. Kite: fixed height $h = 300$ ft. Let x = horizontal distance, L = string length. $L^2 = x^2 + 300^2 = x^2 + 90000$.

Given $\dfrac{dx}{dt} = 25$ ft/s. Find $\dfrac{dL}{dt}$ when $L = 500$.

First find x when $L = 500$: $x^2 = 500^2 - 90000 = 250000 - 90000 = 160000 \Rightarrow x = 400$.

Differentiate:

$$2L\frac{dL}{dt} = 2x\frac{dx}{dt}$$

$$\frac{dL}{dt} = \frac{x}{L}\cdot\frac{dx}{dt} = \frac{400}{500}(25) = \frac{10000}{500} = 20 \text{ ft/s}$$

17. Box: $V = lwh$. At this instant, $l = 4$, $w = 3$, $h = 2$, with $\dfrac{dl}{dt} = 2$, $\dfrac{dw}{dt} = -1$, $\dfrac{dh}{dt} = 3$.

Differentiate using the extended product rule (three factors — each term holds two factors still while differentiating the third):

$$\frac{dV}{dt} = \frac{dl}{dt}\cdot wh + l\cdot\frac{dw}{dt}\cdot h + lw\cdot\frac{dh}{dt}$$

$$\frac{dV}{dt} = (2)(3)(2) + (4)(-1)(2) + (4)(3)(3)$$

$$= 12 + (-8) + 36 = 40$$

Answer: the volume is increasing at a rate of 40 ft³/min.

18. Spotlight: fixed perpendicular distance 50 m. Let x = runner's distance along the track from the closest point, θ = angle between the beam and the perpendicular.

$$\tan\theta = \frac{x}{50}$$

Differentiate with respect to t:

$$\sec^2\theta \cdot \frac{d\theta}{dt} = \frac{1}{50} \cdot \frac{dx}{dt}$$

$$\frac{d\theta}{dt} = \frac{1}{50\sec^2\theta} \cdot \frac{dx}{dt} = \frac{\cos^2\theta}{50} \cdot \frac{dx}{dt}$$

At $x = 50$: $\tan\theta = \frac{50}{50} = 1 \Rightarrow \theta = 45°$, so $\cos\theta = \frac{\sqrt{2}}{2}$, and $\cos^2\theta = \frac{1}{2}$.

Given $\frac{dx}{dt} = 6$ m/s:

$$\frac{d\theta}{dt} = \frac{\frac{1}{2}}{50}(6) = \frac{6}{100} = 0.06 \text{ rad/s}$$

Applied

19. $R = xp$. Given $x = 100$, $p = 20$, $\frac{dx}{dt} = 5$, $\frac{dp}{dt} = -0.5$.

$$\frac{dR}{dt} = \frac{dx}{dt} \cdot p + x \cdot \frac{dp}{dt}$$

$$\frac{dR}{dt} = (5)(20) + (100)(-0.5) = 100 - 50 = 50$$

Answer: revenue is increasing at \$50 per week.

20. $A = \pi r^2$. Given $r = 20$, $\frac{dr}{dt} = 0.5$.

$$\frac{dA}{dt} = 2\pi r \cdot \frac{dr}{dt} = 2\pi(20)(0.5) = 20\pi \text{ m}^2/\text{min}$$

21. $D = \frac{P}{A}$. Given $P = 5000$, $\frac{dP}{dt} = 200$, $A = 25$, $\frac{dA}{dt} = 3$.

Using the quotient rule with respect to t:

$$\frac{dD}{dt} = \frac{\frac{dP}{dt} \cdot A - P \cdot \frac{dA}{dt}}{A^2}$$

$$\frac{dD}{dt} = \frac{(200)(25) - (5000)(3)}{25^2} = \frac{5000 - 15000}{625} = \frac{-10000}{625} = -16$$

Answer: population density is currently decreasing at 16 people per square mile, per year — even though the population itself is growing, the developed area is growing fast enough to outpace it.

22. $V = \frac{4}{3}\pi r^3$. Given $\frac{dV}{dt} = 100$ ft^3/min. Find $\frac{dr}{dt}$ at $r = 5$.

$$\frac{dV}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$$

$$100 = 4\pi(25) \cdot \frac{dr}{dt} = 100\pi \cdot \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{100}{100\pi} = \frac{1}{\pi} \text{ ft/min}$$